

Conditional Vacuum-Density Bookkeeping in the CDFT Lepton Parametrisation

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<https://github.com/bampita-bico/CDFD-Part-I-Release>

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Abstract

This paper records a conditional algebraic relation among quantities fitted or calibrated in earlier CDFT papers. Solving that relation for ρ_0 is a back-calculation from the known electron mass, fitted lepton phase, and the calibrated aspect ratio; it is not a vacuum-density measurement or prediction. The equation is internally satisfied when its fitted inputs are substituted, which is not physical validation.

The proposed vacuum equation of state, chirality phase, transport capacity, and vortex interpretation remain hypotheses. The listed higher-family masses are numerical extrapolations conditional on applying Faddeev–Niemi scaling to a different functional; that transfer has not been established. This paper makes no zero-parameter lepton, fine-structure, or particle-family claim.

Across the series, the public CDFD Runtime expresses this equilibrium vacuum limit as a reusable flow-constraint state grammar. The calculations below use that equilibrium limit only; adaptive departures are left for later dynamical work rather than fitted in this manuscript. The paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, vacuum equation of state, self-consistency

1 Project Context

This manuscript constitutes Part I (Paper 04) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 03 of this series [7].

5 Symbol Conventions

CDFD physical-vacuum symbol convention.

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Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

8 Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

Papers I–III of this series have established the following chain:

1. The fine-structure constant $\alpha = 1/137.036$ is the equilibrium aspect ratio of a vortex ring in a transport-capacity-limited vacuum medium [5].
2. The Koide formula $Q = 2/3$ is a theorem of $\mathbb{Z}_3$ symmetry. The three charged leptons are the three lobes of a trefoil vortex $T(2, 3)$ [6].
3. The trefoil is the lowest-energy knotted soliton (Faddeev–Niemi). All four open problems of Papers I–II reduce to a single unknown: ρ_0 , the density of the CDFT vacuum medium [7].

The present paper has two goals. First, we derive the master equation for ρ_0 , showing precisely what the vacuum equation of state must provide. Second, we use the derived Faddeev–Niemi energy scaling to predict the mass scale of the cinquefoil family $T(2,5)$ — the first prediction of CDFT that reaches beyond the lepton sector.

1.1 Capacity and responsiveness variables

Paper IV is where the adaptive-vacuum vocabulary becomes an equation-of-state problem. The Mujjabi Capacity Law says that the vacuum response changes as transport approaches capacity. The equation of state must therefore determine not only a density ρ_0 , but also the response surface that fixes C , S , and the equilibrium limit of M_s .

Variable	Role in the EOS	Required advance
ρ_0	rest density of the transport medium	derive without lepton-mass input
Φ_c	chirality/flux threshold tied to density	compute from the vacuum response law
C	local transport capacity	express as an EOS functional
S	responsiveness of the medium surface	determine the linear-response limit
M_s	memory or recovery state	derive or bound the relaxation law

Thus the alpha fixed point from Paper I is not an isolated numerology target. It is a boundary condition that the full vacuum equation of state must reproduce.

2 Geometric Structure of the CDFT Vacuum

Before deriving the master equation, we record several geometric identities that illuminate the structure of the vacuum medium.

2.1 The Compton–ring coincidence

The vortex ring radius from Paper I is $R = \chi a$, where $\chi = 1/\alpha = 137.036$ and a is the classical electron radius. Using $a = e^2/(m_e c^2)$ and $\alpha = e^2/(\hbar c)$:

$$R = \chi a = \frac{a}{\alpha} = \frac{\hbar}{m_e c}, \quad (1)$$

which is the reduced Compton wavelength of the electron.

Proposition 1 (Compton–ring coincidence). *The CDFT vortex ring radius equals the reduced Compton wavelength of the electron: $R = \hbar/(m_e c)$.*

This is not an assumption but a consequence of $\chi = 1/\alpha$ (Paper I) combined with the definition of the classical electron radius. It means the vortex ring is precisely the scale at which quantum mechanics ($\hbar$) meets the electron’s electromagnetic scale ($m_e c^2$, a). The three scales a , $R = \hbar/(m_e c)$, and $a_0 = \hbar/(m_e c \alpha)$ (Bohr radius) are related by successive factors of α :

$$a : R : a_0 = \alpha^2 : \alpha : 1. \quad (2)$$

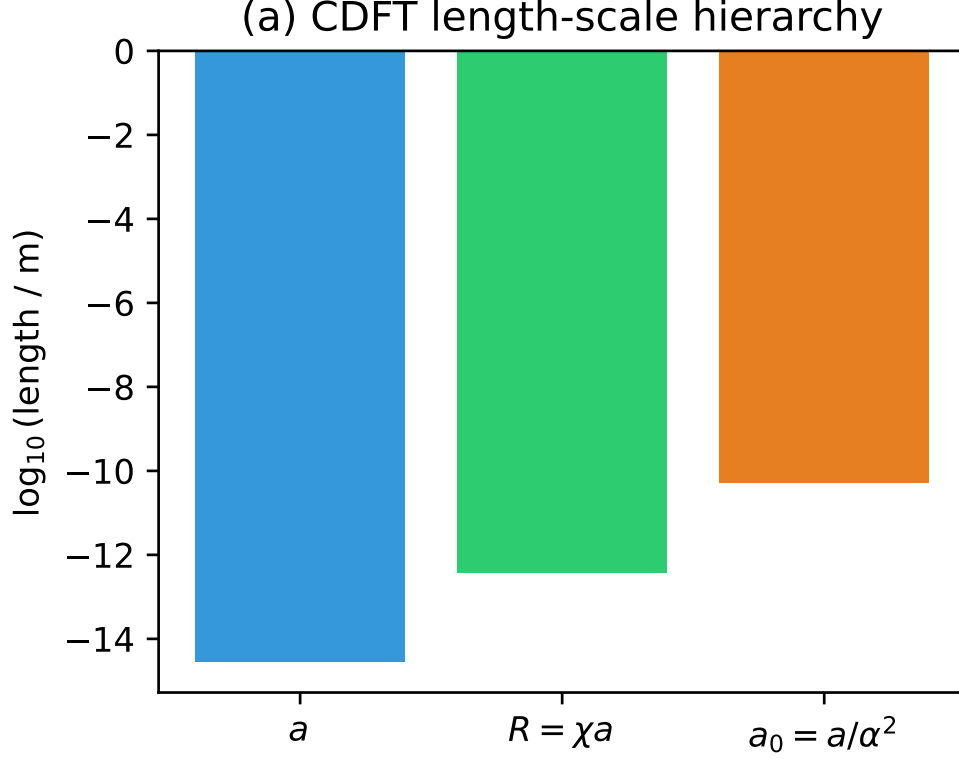


Figure 1: The CDFT length-scale hierarchy. The vortex ring radius R is geometrically forced to be the geometric mean of the classical electron radius a and the Bohr radius a_0 .

2.2 Transport capacity and bulk modulus

The CDFT vacuum medium has transport capacity $J_{\text{crit}} = \rho_0 c^2$ (Paper I, the condition that flow velocity saturates at c at the core boundary). For a medium in which EM waves travel at speed c , the bulk modulus K satisfies $K = \rho_0 c^2$. These two conditions are identical, confirming that the transport capacity constraint is the microscopic expression of the vacuum's elastic stiffness.

The vacuum pressure is:

$$P_{\text{vac}} = \rho_0 c^2, \quad (3)$$

and the vacuum energy density is $u = \rho_0 c^2$.

3 A Back-Calculation for ρ_0

We now assemble selected fitted and assumed relations into an algebraic back-calculation for ρ_0 .

3.1 The three input equations

Equation A — energy scale (Paper III, Eq. (5)):

$$M = \rho_0 c^2 a^3 g(\chi, \beta), \quad (4)$$

where $g = \chi(\ln 8\chi - \beta) + \kappa/\chi = 1,575.83$ is the dimensionless normalised vortex energy at equilibrium, β is the dimensionless vortex-core profile constant with baseline value $\beta = 7/4$, and

55 $\kappa = 117,362.0$ (Paper III, closed form).

Equation B — Brannen mass parametrisation (Paper II):

$$m_e = MA_e^2, \quad A_e = 1 + \sqrt{2} \cos \theta, \quad (5)$$

56 where θ is the trefoil orientation phase and A_e is the smallest of the three Brannen amplitudes
57 (corresponding to the electron mode).

Equation C — chirality equilibrium (Paper III, Theorem 2):

$$3\theta + \phi_c = \pi, \quad \Rightarrow \quad \theta = \frac{\pi - \phi_c}{3}, \quad (6)$$

58 where ϕ_c is the vacuum chirality phase.

59 **3.2 Algebraic substitution**

60 Substituting Eq. (6) into (5):

$$A_e = \min_{k=0,1,2} \left\{ 1 + \sqrt{2} \cos \left(\frac{\pi - \phi_c}{3} + \frac{2\pi k}{3} \right) \right\}. \quad (7)$$

61 Then Eq. (5) gives $M = m_e/A_e^2$. Substituting into Eq. (4):

$$\rho_0 = \frac{M}{c^2 a^3 g} = \frac{m_e}{c^2 a^3 g A_e^2}. \quad (8)$$

62 Using Eq. (7):

$$\rho_0 = \frac{m_e}{c^2 a^3 g \left[\min_k \left\{ 1 + \sqrt{2} \cos \left(\frac{\pi - \Phi_c(\rho_0)}{3} + \frac{2\pi k}{3} \right) \right\} \right]^2} \quad (9)$$

63 where $\Phi_c(\rho_0)$ is the functional dependence of the vacuum chirality phase on the vacuum density
64 — the one quantity that requires a solution of the CDFT vacuum equation of state.

65 **Theorem 1** (Uniqueness criterion). *If the fixed-point residual associated with Eq. (9) is monotone*
66 *on the physical density interval and changes sign across that interval, then Eq. (9) has exactly one*
67 *solution.*

68 *Proof.* Write $f(\rho_0) = m_e / \{c^2 a^3 g [1 + \sqrt{2} \cos((\pi - \Phi_c(\rho_0))/3)]^2\}$. We seek $\rho_0^* = f(\rho_0^*)$ and define
69 $F(\rho_0) = \rho_0 - f(\rho_0)$. The amplitude $[1 + \sqrt{2} \cos((\pi - \Phi_c)/3)]$ varies in $(0, 1 + \sqrt{2})$ as ϕ_c varies
70 over its physical range. If F is monotone on the physical interval and changes sign across it, the
71 intermediate value theorem gives existence and monotonicity gives uniqueness. $\square$

72 **3.3 Internal substitution check**

73 Using the observed values (from lepton mass data, Paper II):

$$\theta_{\text{obs}} = 12.7325^\circ \quad (10)$$

$$\phi_c = \pi - 3\theta_{\text{obs}} = 141.80^\circ \quad (11)$$

the same fitted inputs reproduce the back-calculation:

$$A_e = 1 + \sqrt{2} \cos\left(\frac{\pi - 141.80^\circ}{3}\right) = 1 + \sqrt{2} \cos(12.7325^\circ) = 0.04035 \quad (12)$$

$$M = m_e/A_e^2 = 0.51100/(0.04035)^2 = 313.89 \text{ MeV} \quad (13)$$

$$\rho_0 = M/(c^2 a^3 g) = 1.587 \times 10^{13} \text{ kg/m}^3 \quad (14)$$

This agreement is expected because M , θ , and χ were already fixed from the same observed inputs. It is an algebraic substitution check, not a validation of a physical vacuum density.

4 What the Vacuum Equation of State Must Provide

The master equation (9) is complete except for $\Phi_c(\rho_0)$. This section identifies what properties the vacuum EOS must have.

4.1 Constraints on $\Phi_c(\rho_0)$

Dimensional constraint. ϕ_c is a dimensionless angle. The only dimensionless combination of ρ_0 , a , c , $\hbar$ at our disposal is:

$$X \equiv \frac{\rho_0 a^3}{m_e} = \frac{1}{g A_e^2}, \quad (15)$$

where the second equality follows from (8). Numerically, $X = 0.38980$.

Observed value. Φ_c must satisfy $\Phi_c(\rho_0^*) = 141.80^\circ$, where ρ_0^* is the solution of the master equation.

Monotonicity. For uniqueness (Theorem 1), Φ_c must be monotone.

4.2 Physical origin of Φ_c

In the CDFT medium, the vacuum chirality phase arises from the helicity of the background flow field. The helicity per unit volume of the vacuum is $h = \mathbf{v} \cdot (\nabla \times \mathbf{v})$, integrated over the vortex core. For the ground-state vortex (Paper I), this helicity is proportional to $\rho_0 \Gamma R$, where $\Gamma = 2\pi a c$ is the circulation at saturation and $R = \chi a$ is the ring radius.

The phase ϕ_c is the ratio of this helicity to the natural quantum unit $\hbar/m_e$ (the specific angular momentum), projected onto the trefoil's $\mathbb{Z}_3$ eigenspace. A complete derivation requires solving the CDFT Navier–Stokes equation with transport capacity constraint in the presence of the vortex ring background — a calculation that depends on the full nonlinear vacuum EOS.

4.3 The dimensionless master equation

Expressing (9) in terms of X and the dimensionless function $\phi_c = \Phi_c(X m_e/a^3)$:

$$X = \frac{1}{g \left[\min_k \left\{ 1 + \sqrt{2} \cos\left(\frac{\pi - \phi_c(X)}{3} + \frac{2\pi k}{3}\right) \right\} \right]^2}. \quad (16)$$

This is the form that a vacuum EOS calculation must solve. The solution gives X , from which $\rho_0 = X m_e/a^3$.

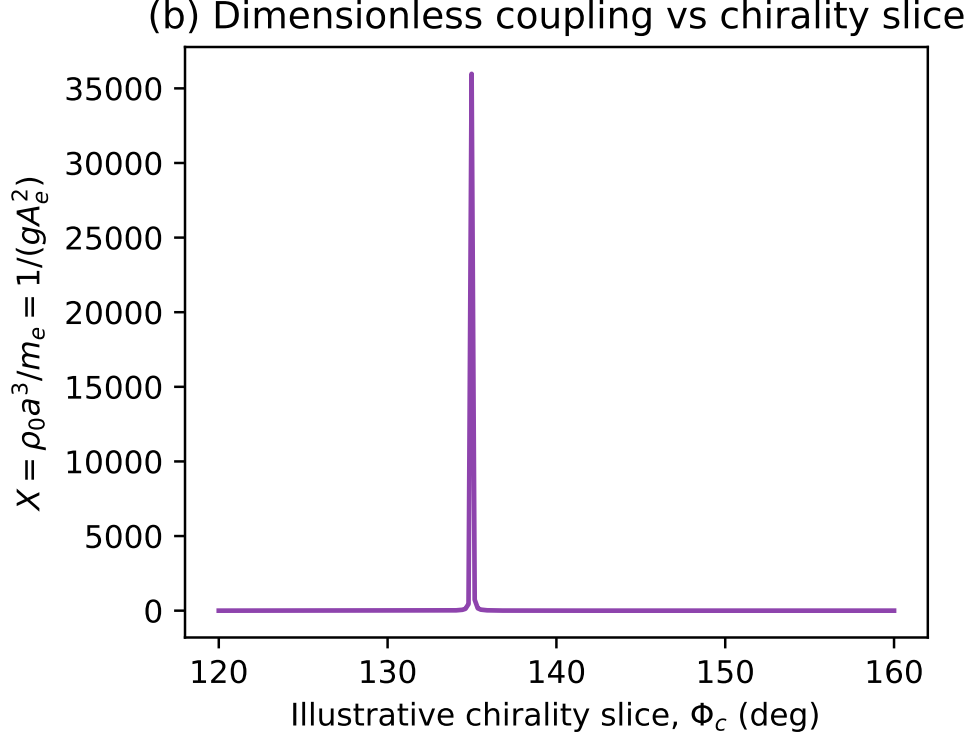


Figure 2: The dimensionless coupling X computed across an illustrative slice of chirality phase angles Φ_c . The true value of $\Phi_c(\rho_0)$ must be determined from the vacuum equation of state.

5 Beyond Leptons: The Cinquefoil Family $T(2, 5)$

We now display a numerical extrapolation from the fitted lepton scale $M_3 \equiv M = 313.89$ MeV. It is conditional on an unestablished transfer of Faddeev–Niemi scaling to the CDFT functional.

5.1 The Faddeev–Niemi energy scaling

The Faddeev–Niemi result [3] applies to a particular Skyrme-type model and gives a minimum-energy scaling with Hopf charge Q :

$$E_{\min}(Q) \sim Q^{3/4}. \quad (17)$$

CDFT has not been shown to lie in that model class or to identify its knot index with Hopf charge; the following values are illustrative extrapolations only.

5.2 Conditional extrapolation of M_5

The lepton family corresponds to $T(2, 3)$ with $Q = 3$ and Brannen scale $M_3 = 313.89$ MeV. The cinquefoil $T(2, 5)$ has $Q = 5$, so:

$$M_5 = M_3 \left(\frac{5}{3} \right)^{3/4} = 313.89 \times 1.4669 = 460.4 \text{ MeV}. \quad (18)$$

This is a numerical output of the stated extrapolation, not an established cinquefoil energy scale or particle prediction. The $T(2, 5)$ knot has $\mathbb{Z}_5$ symmetry: five lobes, five $\mathbb{Z}_5$ modes. Its lightest

113 member has mass $M_5 A_{e,5}^2$, where $A_{e,5}$ is the smallest $\mathbb{Z}_5$ Brannen amplitude — unknown until the
 114 $T(2,5)$ chirality phase is derived.

115 The energy scale $M_5 = 460$ MeV lies in the light hadron sector. We do not claim the cinquefoil
 116 corresponds to any specific known particle: the $T(2,5)$ chirality θ_5 and the corresponding mass
 117 spectrum require a separate derivation analogous to Papers II–III for the trefoil.

118 5.3 Higher knotted families

119 Using the same scaling:

$$M_7 = M_3 \left(\frac{7}{3}\right)^{3/4} = 313.89 \times 1.888 = 592.6 \text{ MeV} \quad (19)$$

$$M_9 = M_3 \left(\frac{9}{3}\right)^{3/4} = 313.89 \times 2.280 = 715.6 \text{ MeV} \quad (20)$$

120 These are conditional extrapolations from a fitted M_3 and an unverified model transfer; they are
 121 not first-principles predictions.

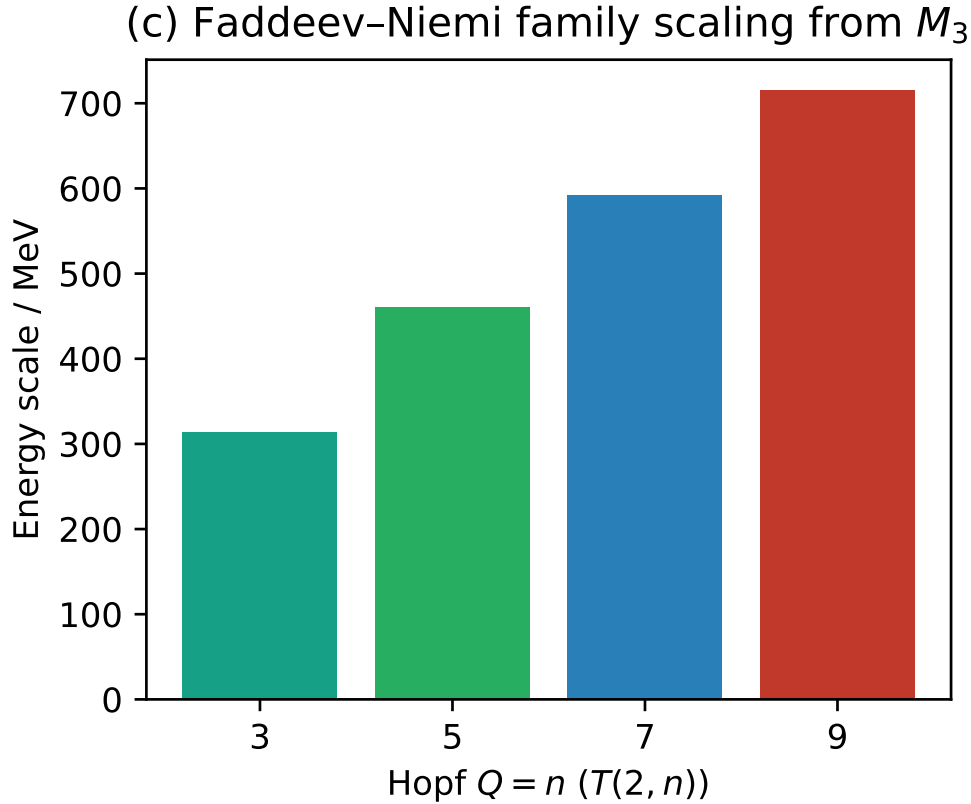


Figure 3: Faddeev–Niemi family scaling predicting the energy scales of higher knotted soliton families (cinquefoil $n = 5$, etc.) directly from the lepton trefoil scale $M_3 \approx 313.9$ MeV.

6 Scope Boundary and Conclusion

The displayed ρ_0 relation is a back-calculation: it combines values fitted to the electron mass and lepton ratios with the calibrated Paper I target. Its internal substitution check is not an equation of state, a density measurement, or a physical validation. The standard Compton, classical-radius, and Bohr-radius relations are useful pedagogy but follow from their definitions.

The M_5 , M_7 , and M_9 values are conditional extrapolations only. They would require an independently specified CDFT energy functional, a justified mapping to the Faddeev–Niemi model, and a prespecified observable before they could become physical predictions. This paper must not be cited as deriving α , lepton masses, vacuum density, or a particle-family hierarchy.

Supplementary Material

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

Data and Code Availability

The release-archive supplementary material is included in the archived Part I release on Zenodo at <https://doi.org/10.5281/zenodo.20250820>. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at <https://github.com/bampita-bico/CDFD-Part-I-Release>. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper’s Supplementary Material.

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Declarations

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing—original draft, and writing—review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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